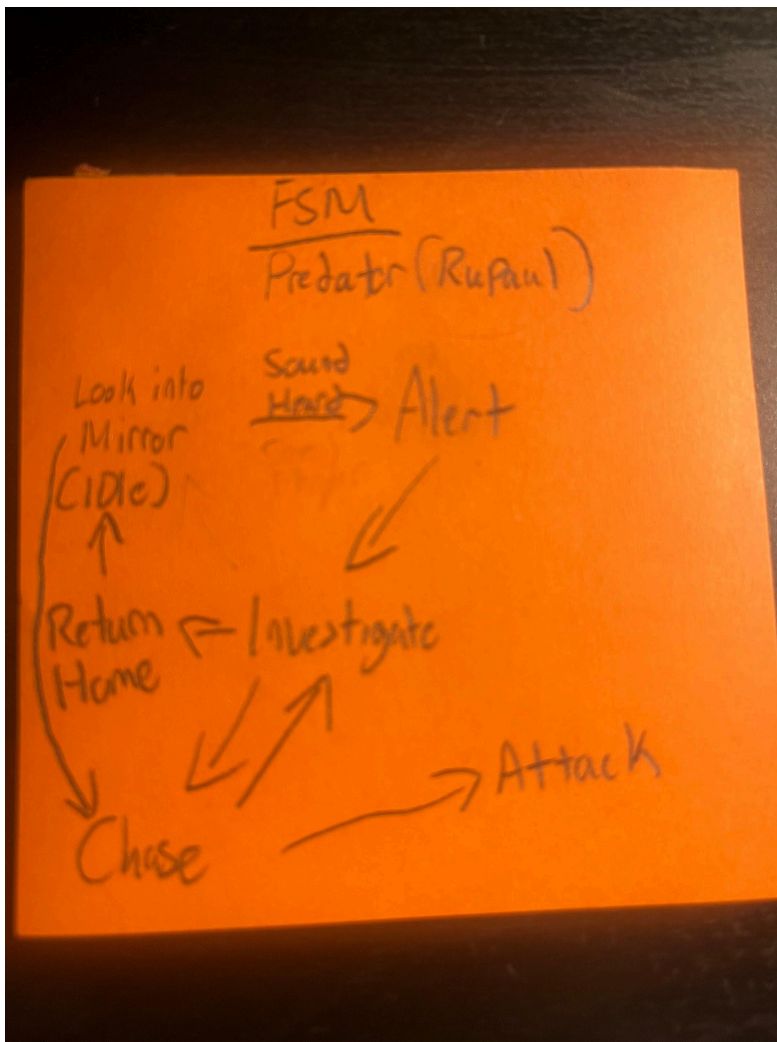


A3 Video Link: <https://youtu.be/eRmvopqgrvM>

Early drafts of the FSM and what all the states did:



Idle: RuPaul's base location. It is here where Rupaul Returns to.

Alert: When Ru hears a sound, Ru will head to the location of the sound.

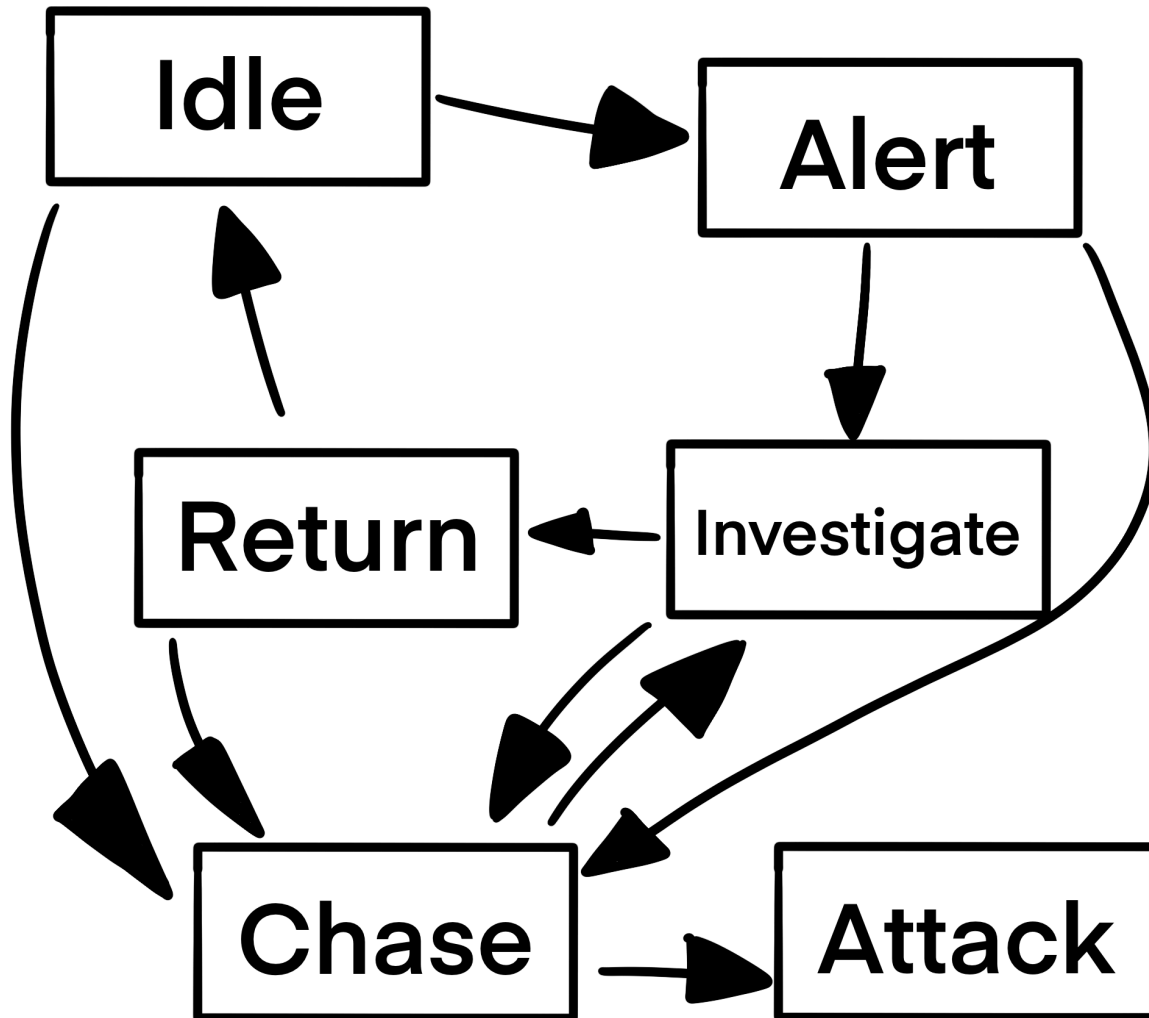
Investigate: Ru searches the location of the sound, or where Ru last saw the Player.

Return Home: Ru returns to Home Base.

Chase: Ru chase the Player.

Attack: Ru kills the Player.

Complete Version:



State Behaviours & Transitions:

Idle: While in Idle RuPaul stands in the center of the map, waiting for noise. RuPaul leaves the Idle state in one of two ways, When she hears a sound which will send her into the Alert state or when she sees the player which will send her into the Chase State.

Alert: While in the Alert state, RuPaul will head towards the location of the sound. Alert can only be entered from the idle state. If RuPaul see the player on the way to the sound location, she will enter the Chase State, Otherwise, when she reaches the sound location she will enter the Investigate state.

Investigate: While in the Investigate state, RuPaul will spin around checking the surrounding area. There are two ways for RuPaul to enter this state, either by reaching the spot where a

sound was last heard or by losing track of the player and checking the last place she saw them. If RuPaul doesn't find the player in this state, she will return to her home base. If she does she enters Chase.

Return: RuPaul will only enter the return state if she found nothing during Investigate. During return, RuPaul will head back to her home base in the center of the map. RuPaul can exit this state in one of two ways, either by reaching home base and entering the Idle state, or by spotting the player on the way back and enter Chase.

Chase: If RuPaul sees the play in any of her other states (with the exception of attack), she will enter chase. While in Chase, RuPaul will quickly move towards the player, if RuPaul loses sight of the player she will enter the Investigation State. If RuPaul gets too close to the player she will enter the attack state.

Attack: When RuPaul gets close to the player, she will enter the attack state, while in the attack state RuPaul will send the player to game over.

Challenges:

I had so much fun making this assignment. As for challenges I faced during this assignment they were very minor and I was able to figure them out eventually with little stress. Some of these challenges were, RuPaul not switching from Alert to Investigate, RuPaul getting stuck on the walls, RuPaul not leaving the Investigate State, RuPaul not entering attack state, and RuPaul completely freezing if hitting a wall in chase. A lot of the solutions to these challenges was switching a $>/<$ or changing the distance in which RuPaul actually switches states. As for RuPaul not killing the player it was very inconsistent, I found that it was due to the speed of RuPaul so to fix it I would just have to slow RuPaul down but I also found it way more enjoyable to have RuPaul speed around in the level. My favourite part of making this level was implementing the 2 sound clips of RuPaul. The first clip plays when RuPaul enters the Alert state and it's RuPaul screaming. The second clip plays one game over and it's RuPaul laughing pitched down.